

Explainable AI for Academic Integrity Verification

Turning a detection flag into a conversation a lecturer can defend

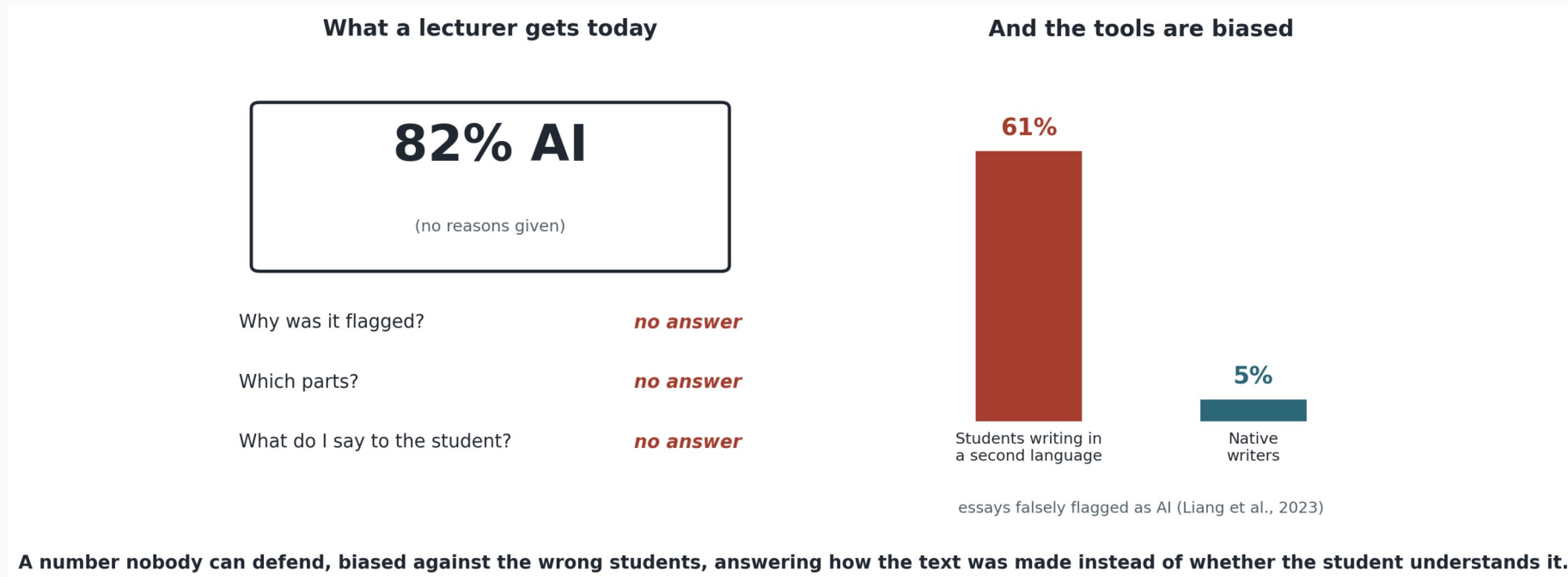
Mykhailo Shpyl

Student number [STUDENT NUMBER]

MSc Artificial Intelligence and Big Data Analytics

Supervisor: Dr Vini Vijayan | Atlantic Technological University

Why this problem, and why this way



A lecturer is given a percentage and nothing else. She cannot defend it if the student challenges it, and it tells her nothing about whether the student understands the work.

It is also unfair. Liang et al. found seven commercial detectors flagged 61% of essays by non-native English writers against about 5% for native writers.

So I did not try to build a better percentage. I asked what a lecturer actually needs, which is evidence and a conversation.

Where the field is now, and where it stops

DETECTING AI TEXT

Strong and improving. DetectGPT needs no training data (Mitchell et al., 2023); SemEval-2024 Task 8 benchmarks across generators. Stops at a score with no reason, and a paraphraser drops DetectGPT from 70.3% to 4.6% (Krishna et al., 2023).

EXPLAINING A MODEL

Attribution methods are established, and so are ways to test whether an explanation is faithful rather than merely plausible (ERASER, DeYoung et al., 2020). Rarely run on detectors.

ARGUMENT MINING

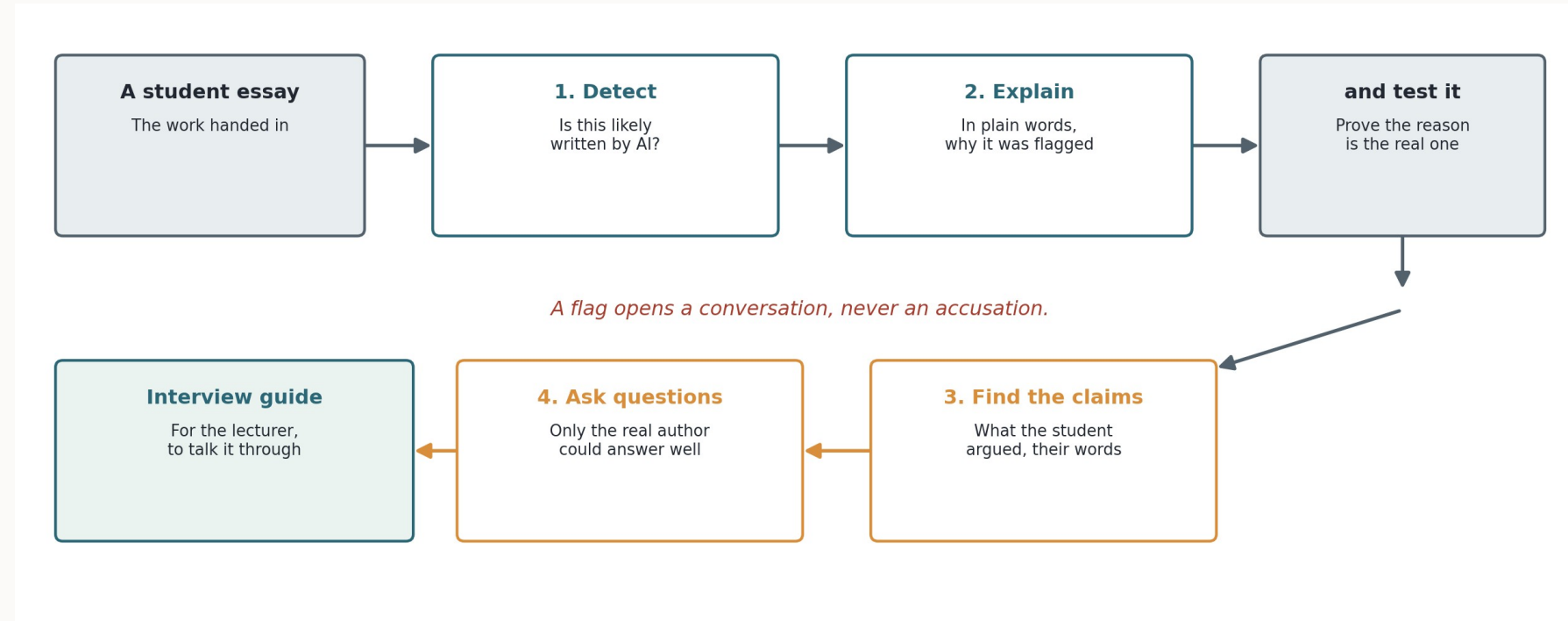
Claims and premises can be recovered from student essays with exact provenance (Stab and Gurevych, 2017). Not connected to integrity work.

GENERATING QUESTIONS

Maturing, with reference-free evaluation (RQUGE, Mohammadshahi et al., 2023) and Bloom's taxonomy for cognitive level. Not grounded in one student's own text.

Each strand works. None of them joins up. This project is the join: a flag carried through to questions drawn from the student's own claims.

One pipeline, five stages, six trained models



Why local: a student submission is personal data. Every model runs in one process on an 8 GB laptop, so nothing is posted to an external API to be judged.

Why six models, not one prompt: each stage is separately trained and separately evaluated, so a failure in one is visible instead of hidden inside a wrapper.

IMPLEMENTATION

The system running, end to end

Verification Interview Guide● hybrid detector ready
RUNS LOCALLY, ON THIS MACHINE

as a gentle but persistent presence that resides within us. In contrast to the poetic world of Dickinson's work, Shakespeare's Othello features a dramatic scene that showcases the tragic hero's downfall. Act 4, Scene 1 is one of the most pivotal scenes in the play, where Iago manipulates Othello into believing his wife Desdemona has been unfaithful. The dialogue between the two characters is characterized by its intensity and conviction: "Villain, be sure thou prove my love a wanton. / 'Tis but a little time till I depart"

Analysing...

Load an AI-written example

Load a real student example

Compare two submissions

2066 words

Claims

Questions each

Question writer

STEP 1 Is this likely AI-written?2.83s

0.96
probability of AI

Flagged as likely AI-generated
transformer 0.9996
style plus perplexity 0.9985

How this happens: two independent readers score the text. One is a language model fine-tuned on matched human and AI essays; the other measures writing habits with formulas anyone can re-run. A small combiner turns the pair into one probability, and it only reads high when both agree. There is no score of 1.0 on this scale: the model is never certain, only confident.

STEP 2 Why, in named writing habits

Live capture: paste a submission, press one button. Verdict, reasons, sentence evidence, claims and questions. Served from 127.0.0.1 with the network cable out.

What the numbers say, and what they do not

0.99

Detector F1, in domain, on 200 held-out essays

0.97

Across six generators it was never trained on

79%

Human abstracts falsely flagged out of domain

24 of 29

Essays where the local 3B model beat the commercial tier, $p = 0.0003$

Significance: 0.99 in domain is the least interesting number here, because one corpus paired with one generator is an easy problem. 0.97 across six unseen generators is the one that carries weight.

Compared with the literature: Liang et al. report 61% false flags on non-native writers. On my in-domain test set the hybrid produces zero false accusations in either group.

The honest number is the red one. Out of domain the detector accuses innocent people, which is why the pipeline continues into evidence and questions instead of stopping at a score.

Three results I had to retract, and one machine

THE DETECTOR THAT READ FORMATTING

My first detector scored a perfect 100%. A rule using only leftover corpus markup reached 92.5% on its own, so the model was reading how the corpus was built. I wrote a cleaning step for both classes and retrained. A function-words-only model still reaches 99.5%, which is the check that what remains is style.

THE FINE-TUNE THAT GAMED MY OWN METRIC

A QLoRA fine-tune scored beautifully on discrimination. Reading its output showed degenerate multiple-choice stems. The score was real and the result was worthless, so it is reported as a failure and a well-formedness gate now sits in front of it.

THE JUDGES THAT FAILED VALIDATION

Three commercial LLMs rated the questions. Cross-model agreement was Krippendorff's alpha of -0.253, worse than chance, and they praised the degenerate output. Reliability without validity, so the judge-free measure is what I report.

8 GB, AND NO HPC

The compute I planned for was withdrawn mid-project. Fit probes measured 8B QLoRA spilling into system RAM at 214 s/step against 2.8 s/step for a 3B. The scope change was made on that evidence and signed off, not on a hunch.

LIMITATIONS

What this does not do, and what it costs

No students were involved. I can show the questions discriminate between a model that has read the essay and one that has not. I cannot yet show they discriminate between students. That was a deliberate scope decision to stay clear of ethics approval, and the classroom study is designed and ready.

Out of domain it is not deployable. 79% false flags on academic abstracts, 61% after fusion. I report that as a failure, not a fix. It needs per-domain calibration and an abstain band before anyone points it at real work.

One corpus, one generator for training. The 0.97 transfer result is reassuring but it is still British academic writing and one family of models.

The commercial comparison is against a free tier, not a frontier model. The claim is narrow and I state it narrowly.

Impact: this is why the system is designed to support a conversation rather than issue a verdict. The limitations are the reason for the architecture, not an apology for it.

How I judged the work, and how I judged myself

EVALUATING THE RESEARCH

The primary measure needs no human and no LLM judge: a question is good if a model that has read the source answers it better than one that has not. That is decided by the source text, so it cannot be gamed by how a question looks. When three LLM judges disagreed with it, I validated the judges three ways, they failed all three, and I kept the measure.

EVALUATING MYSELF

What I would do differently: I would build the audit before the first result rather than after it. All three retractions came from checks I ran late. I also under-estimated how much of an MSc is deciding what not to build; the scope was cut twice and both cuts improved it.

The rule I set at the start, and kept: never trust a score you cannot explain, and never trust one you have not tried to break. It cost me three results I had already written down as wins. That is the single most useful thing this year taught me, and it is why I would defend the evaluation method as the contribution I would keep if I could only keep one.

Where this goes next

The classroom study. The guide is a ready-made instrument: flagged essays, real students, and whether the questions distinguish authorship in practice rather than in simulation.

Per-domain calibration and an abstain band, so the out-of-domain failure is handled rather than reported.

A three-class variant adding partial-AI, which was deliberately left out of this phase because it overlaps heavily with both other classes.

Do not accuse a student with a number nobody can defend. Give the lecturer the evidence and the questions, and let the student answer for their own work.

Everything is public and reproducible:

github.com/MichaelShpyl/AI_research

Thank you. I am happy to answer any questions.